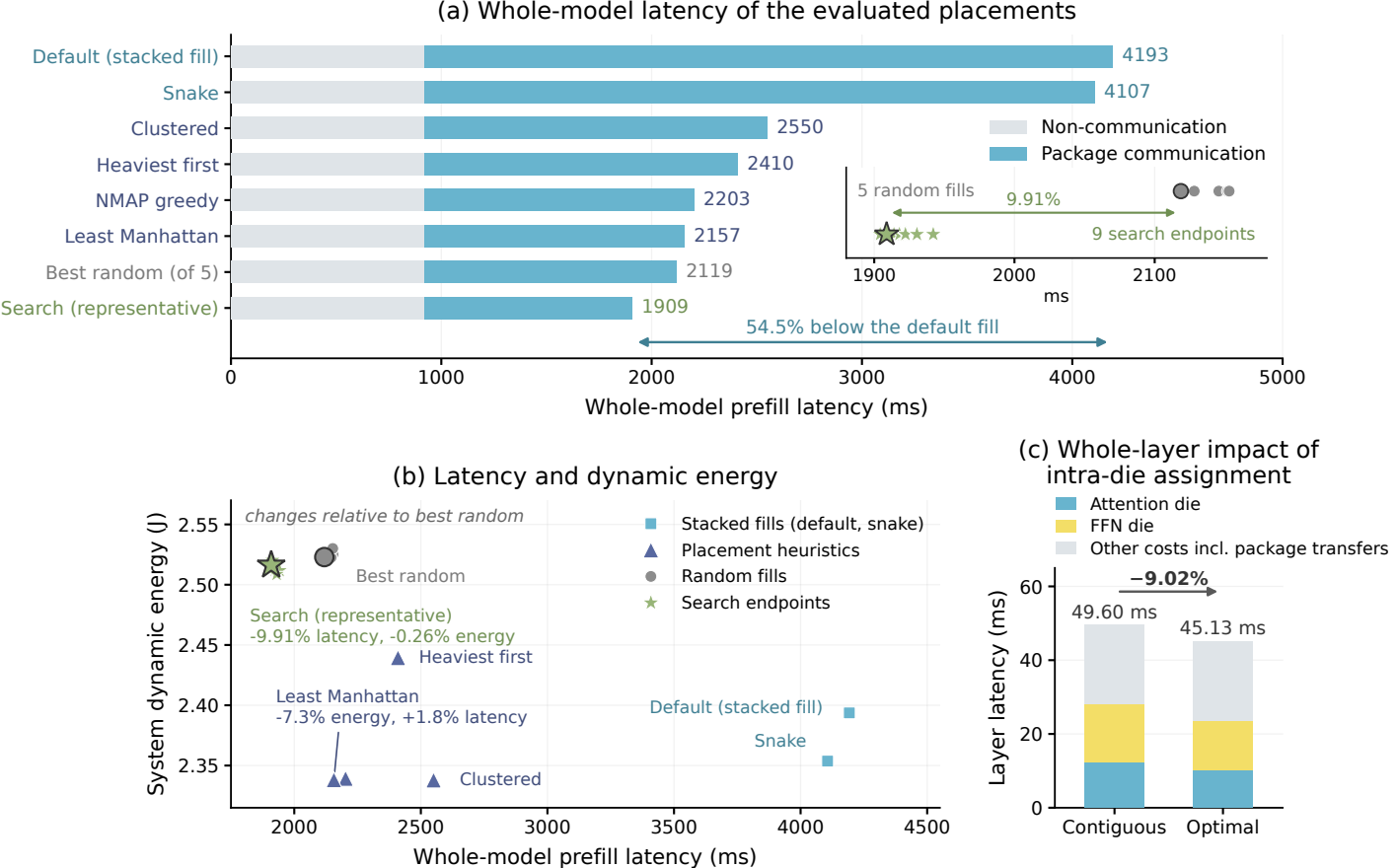
(a) Whole-model latency of the evaluated placements
Default (stacked fill)
Snake
Clustered
Heaviest first
NMAP greedy
Least Manhattan
Best random (of 5)
Search (representative)
4193
4107
2550
2410
2203
2157
2119
1909
Non-communication
Package communication
5 random fills
9.91%
9 search endpoints
1900
2000
2100
ms
54.5% below the default fill
0
1000
2000
3000
4000
5000
Whole-model prefill latency (ms)
(b) Latency and dynamic energy
changes relative to best random
Best random
Search (representative)
-9.91% latency, -0.26% energy
Heaviest first
Least Manhattan
-7.3% energy, +1.8% latency
Clustered
Default (stacked fill)
Snake
Stacked fills (default, snake)
Placement heuristics
Random fills
Search endpoints
System dynamic energy (J)
2.55
2.50
2.45
2.40
2.35
2000
2500
3000
3500
4000
4500
Whole-model prefill latency (ms)
(c) Whole-layer impact of intra-die assignment
Attention die
FFN die
Other costs incl. package transfers
−9.02%
49.60 ms
45.13 ms
Layer latency (ms)
60
40
20
0
Contiguous
Optimal